

Graphene growth by Chemical Vapor Deposition

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B. Tech in PHYSICS

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Abstract

The chemical vapor deposition (CVD) growth of graphene has become one of the most promising approaches for producing large-area, high-quality two-dimensional (2D) materials suitable for advanced electronic, optical, and mechanical applications. Unlike exfoliated graphene, CVD-grown graphene offers controllable layer thickness, uniformity, and scalability, making it compatible with industrial processing. This work investigates the growth of graphene on catalytic metal substrates, focusing on methane-based CVD synthesis and the influence of process parameters such as gas flow rates, temperature, annealing duration, and cooling conditions on the resulting film quality. Key challenges in high-quality graphene fabrication are examined, including defect formation, grain boundary evolution, and transfer-related degradation. Special emphasis is placed on comparing PMMA-assisted and PET/silicone-assisted transfer techniques onto SiO₂/Si substrates, highlighting how factors such as etching duration, film support, and cooling rate affect the structural integrity of the transferred graphene. Raman spectroscopy and optical microscopy are employed to evaluate layer number, crystalline quality, and defect density, while identifying signatures of bilayer graphene and regions of incomplete transfer. Finally, the study outlines the potential of optimized CVD-grown graphene for applications in flexible electronics, transparent conductors, sensors, and future 2D material heterostructures. The results emphasize the importance of controlled growth conditions and improved transfer techniques for enabling reliable, scalable graphene technologies.

Chapter-1

Introduction

1.1 What are 2D materials

Two-dimensional (2D) materials are crystalline materials consisting of a single layer (or a few atomic layers) of atoms.

In the recent decades, 2D materials have gained notable attention due to their extraordinary structural and Physicochemical attributes. These materials characterized by their two-dimensional arrangements of atoms, have emerged as versatile tools with application spanning a magnitude of disciplines, captivating researchers. This fascination can be traced back to the groundbreaking synthesis of graphene in 2004, a pivotal moment that sparked a surge of interest and launched investigations into novel 2D materials. Inspired by graphene's performance, scientists uncovered, designed and engineered a diverse array of 2D materials. (Like: silicone, hexagonal boron nitride, black phosphorus, metal organic framework, covalent organic framework, MXenes, and MBenes.) [3] , Researchers have investigated the fundamental mechanisms governing the behavior of 2D materials, with an emphasis on the intricate relationships between structure, property, and performance. 2D materials have emerged as frontrunners in a range of applications, due to their high surface-to-volume ratio, good mechanical properties, controllable electrical and thermal conductivities and , and chemical tunability from revolutionizing energy storage and harvesting technologies to catalyzing advancements in (photon)-catalysis and water purification, 2D materials impact across an array of fields.

1.2 Graphene

Graphene, a single layer of graphite, has been one of the first two-dimensional material (one atom thick) to be isolated in 2004 by Andre Geim and Konstantin Novoselov at the University of Manchester. Although, Graphite has been known since the sixteenth century and has been widely used in industry for steel making, as barke lining or as dry lubricant in some other devices. But it was not until 1985 when the discovery of fullerenes greatly expanded the number of known carbons

allotropes and furthermore suggested the existence of its 1D form, carbon nanotubes, which were first demonstrated in 1991. But the theoretical studies of graphite, fullerenes and nanotube were performed using a single layer of graphite as starting material after 2004.

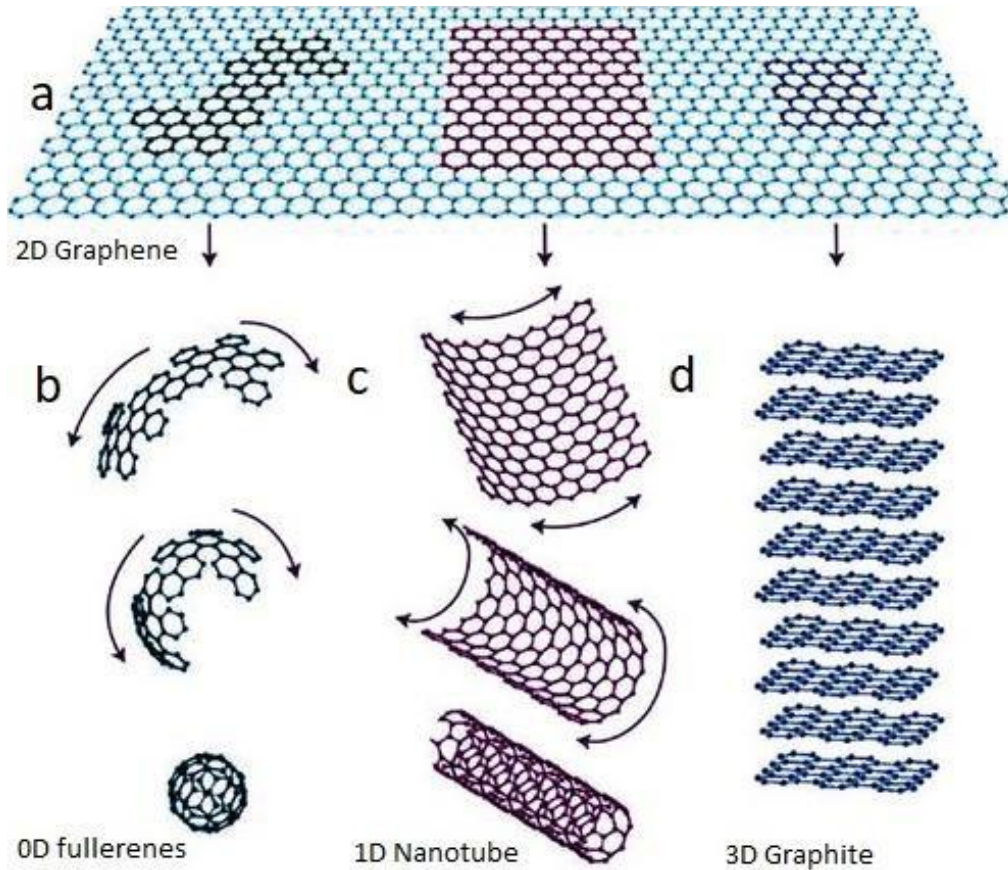


Figure 1: a) 2d graphene which is obtain by exfoliating graphite, (b) 0d fullerenes, (c) 1d carbon nanotube, (d) 3d graphite which is obtain by stalking graphene [6].

Graphene is an atomically thin Sp^2 carbon layered material, which is covalently bounded by a honeycomb lattice and their building block are attracted by weak Van-der waals force [5]. It can be wrapped up into 0 dimension. It is the most stable material that has ever been found with remarkable physical properties like high carrier mobility (approximately $20,000 \text{ cm}^2/\text{V}\cdot\text{sec.}$), electrical and thermal conductivity and high optical

transparency (absorb 2.3% of visible light) [6].

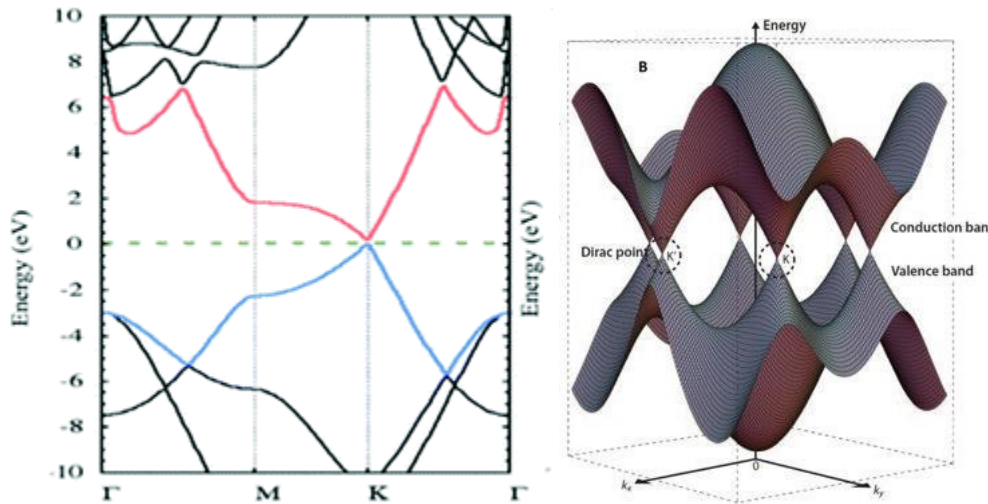


Figure 2: Represent conduction and valance band of graphene at Dirac Fermi level [7] .

However, due to its lack of a band gap (Ref: Figure 2) graphene is a Semi-metal and is hence not appropriate for use as an electrical switch. Band gap engineering is required to open a band gap in graphene without compromising any of its other properties. The methods and procedures that have been reported the most frequently are doping, functionalization, faulty or hybrid structures, substrate driven band gap opening, and quantum confinement. Graphene is presently being developed for a wide range of applications, and many more have been suggested [1]. Among these are solar cells, electric circuits, flexible display screens, and pharmaceutical, medical, and industrial processes that could profit from new graphene materials [6,7].

Chapter-2

Synthesis Techniques

2.1 Brief Introduction of Synthesis Techniques

It is true that first single layer or few layers isolation technique was top-to-bottom exfoliation method. But currently there exist a wide range of techniques to produce 2D materials, use of these techniques depends on the required applications. The most common synthesis methods include bottom-up approaches, such as chemical vapor deposition (CVD), chemical vapor transport (CVT) or other physical deposition methods. On the other hand, top-to-bottom approaches are Mechanical exfoliation, Liquid phase exfoliation and Sputtering [11].

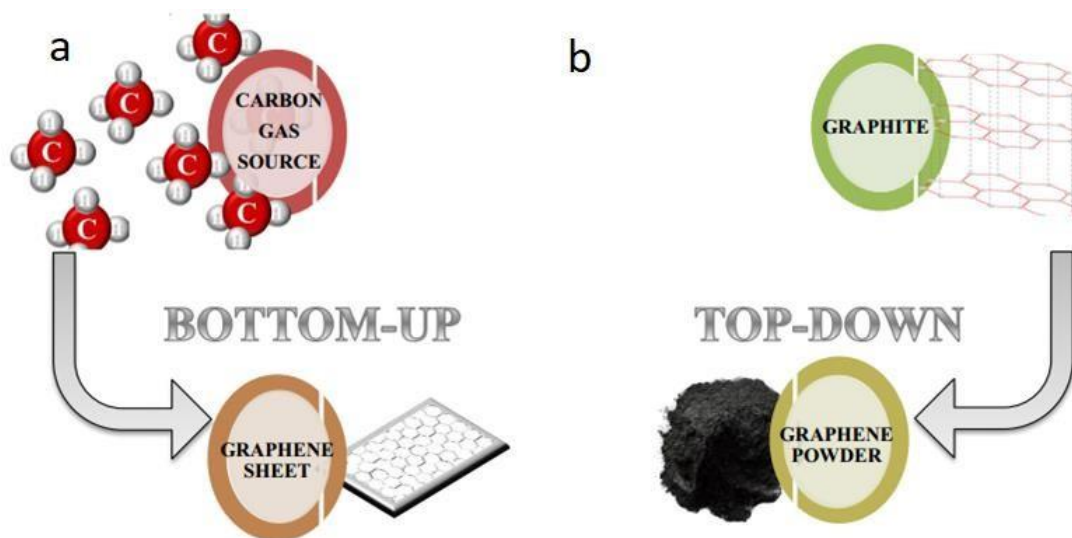


Figure 3: (a) Represent bottom- up approach where we use carbon gas to get thin layer. (b) represent top-down approach where we use bulk carbon to get thin layer.

The top-down approach involves breaking down bulk layered crystals into thinner sheets using different methods. These techniques typically yield high-quality crystalline flakes but with limited control over size, shape, and thickness, making scalability a major challenge. In contrast, the bottom-up approach constructs 2D materials atom-by-atom or molecule-by-molecule using various techniques.

Bottom-up methods offer better control over material properties, such as thickness uniformity and domain size, and are more suitable for large-scale production. In this thesis I will give an overview about chemical vapor deposition and mechanical exfoliation [1].

2.2 Chemical Vapor Deposition

Since 2004, Graphene has attracted enormous interest because of its unique properties. Chemical vapor deposition (CVD) has emerged as an important method for the preparation and production of 2D materials for various applications, since the method was first reported in year between 2008 and 2009.

It is widely used technique for the fabrication of uniformly thin films and nanostructures on various substrates in large area. In this process, volatile precursor gases (CH_4 , H_2 , MoS_3 or WO_3) react or decompose on the heated surface of a substrate (Cu, Ni, etc.) to form a solid material, while by-products are typically removed by gas flow. CVD offers precise control over the material composition, thickness, and morphology, making it an essential method for producing semiconductors, optical coatings, and two-dimensional materials such as graphene and transition metal dichalcogenides (TMDCs) [[38], [39]].

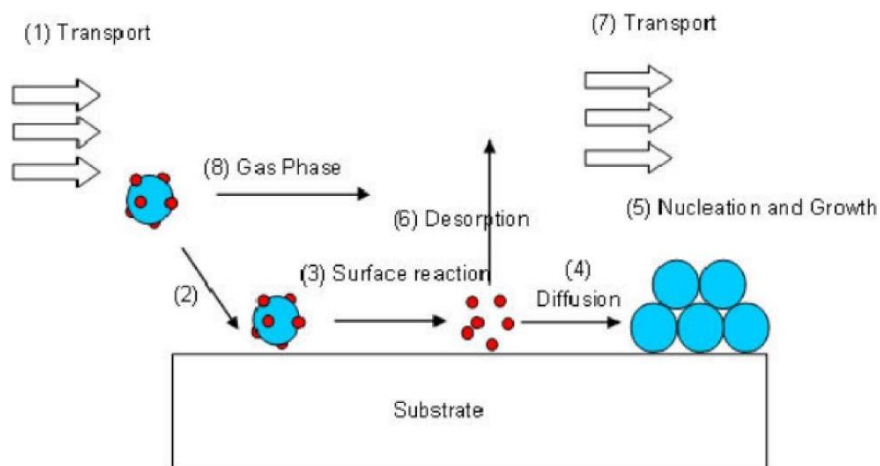


Figure 4: Schematic representation of CVD process

Furthermore, this approach is very scalable since it is easy to control the size and thickness of the 2D flake by modifying several growth factors like temperature, gas flow rate, and pressure. These features make this process a strong candidate for the

large-scale synthesis of 2D materials for use in devices and the industrial sector [13]. The production of 2D materials can be accomplished by a broad range of CVD techniques, including Metal-Organic Chemical Vapor Deposition (MOCVD), Low Pressure Chemical Vapor Deposition (LPCVD), and Plasma Enhanced Chemical Vapor Deposition (PLCVD) [12].

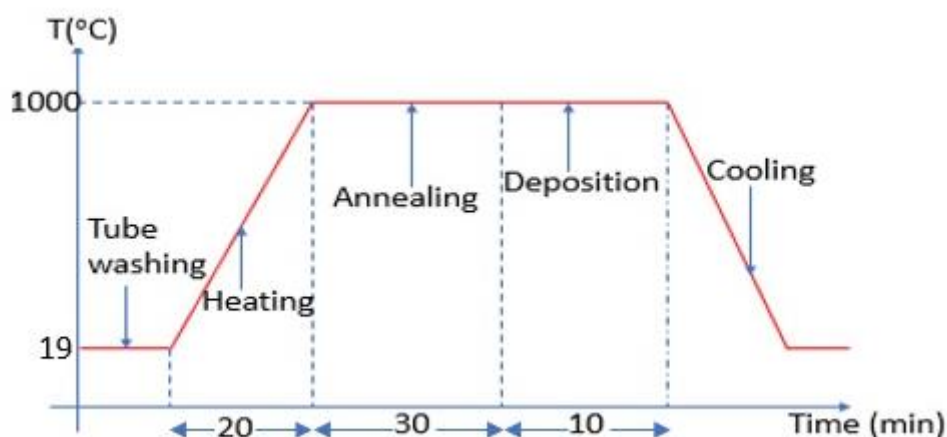


Figure 5: Time-line for the CVD process

Following annealing, the graphene growth stage begins when methane (CH_4) and hydrogen (H_2) are introduced into the furnace. CH_4 decomposes on the hot Cu surface, allowing carbon atoms to diffuse and nucleate into hexagonal graphene domains. The amount of methane is a critical parameter; as shown in your results, low CH_4 flows (15 and 25 sccm) do not produce prominent Raman G or 2D peaks, while **45 sccm** enables bilayer graphene formation. Hydrogen acts as both a catalyst and a cleaning agent by removing weakly bonded carbon and controlling nucleation density. The interplay between CH_4 concentration, H_2 flow, temperature, and pressure governs graphene thickness, uniformity, and crystal size. Once the Cu surface becomes fully covered, growth naturally saturates due to the self-limiting nature of carbon dissolution on Cu.

The final stage is cooling, which has a significant impact on the structural integrity of graphene. **Fast cooling**—as used in your successful samples—helps preserve domain structure, minimizes thermal stress, and maintains bilayer stacking with fewer defects. It prevents wrinkle formation, excessive grain boundary growth, and undesired etching effects. In contrast, **slow cooling** can lead to structural degradation such as crack formation, wrinkling, and even loss of the first graphene underlayer, as observed in your PMMA-transferred sample where long etching times caused

incomplete coverage. Combined, these steps highlight why CVD is such a powerful method for synthesizing 2D materials: it enables scalable, large-area production of uniform films with controlled thickness, low defect density, and tunable morphology. Its compatibility with process optimization—through substrate treatment, gas flow regulation, annealing, and cooling control—makes CVD indispensable for fabricating graphene for sensors, optoelectronic devices, transparent electrodes, and advanced nanoelectronics applications.

2.3 Micromechanical Exfoliation

Micromechanical exfoliation is one of the earliest and most effective methods for isolating atomically thin layers of materials from their bulk counterparts. This method made it feasible for Novoselov and Geim to discover graphene in 2004 [14]. They also used this method to create BH, MoS₂, NbSe₂, and Bi₂Sr₂CaCu₂O_x, among other layered bulk materials, into a variety of single-layer 2D crystal [1],[15]. Although micromechanical exfoliation is limited by scalability challenges, but it produces nearly defect-free interfaces and high-quality thin layer.

Isolation of single or few layers from a bulk material stuck to an adhesive tape piece is made possible by weak van der Waals interaction between individual atomic layer. Large numbers of 2D flakes are produced when the bulk layered material is repeatedly cleaved with the aid of the adhesive tape. These flakes can then be transferred to rigid substrates like *Si/SiO₂* or flexible polymeric films by carefully pressing the tape up against the substrate's surface and slowly peeling it off. The produced 2D nanosheets crystal structure is identical to that of their bulk counterpart [16].

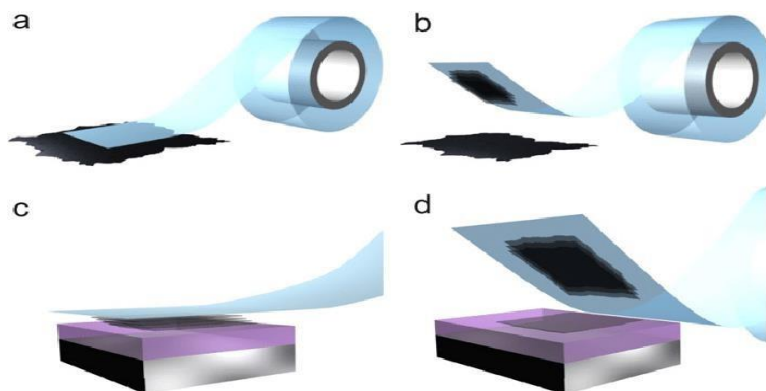


Figure 6: Micromechanical exfoliation of 2D crystals. (a) Adhesive tape is pressed against a 2D crystal so that the top few layers are attached to the tape (b). (c) The tape with crystals of layered material is pressed against a surface of choice. (d) Upon peeling off, the bottom layer is left on the substrate [17].

The yield of mono-layer and few-layer thick samples that are deposited is quite poor, even if this approach of producing 2D materials is easier to use, faster, and less expensive to run than other current methods. Additionally, it has been noted that the samples produced by this process have an incredibly small effective size (about $5\mu\text{m}$)[18], [59]. Some modifications to the current process, which are discussed below, can help to somewhat reduce these issues.

i. Micromechanical Hot Exfoliation

Micromechanical hot exfoliation is an alternative method to improve the yield, size and quality of exfoliated flakes. This technique was purposed by Yuan Huang et al [20]. In this technique, we used to isolate thin layers of materials, particularly 2D crystals, by applying both mechanical force and elevated temperatures.

"Modified Scotch Tape Exfoliation Method," or "MHE," Unlike the MME technique, this method uses oxygen plasma treated silicon wafers as substrates for the deposit of high-quality 2D materials. The silicon wafers are cleaned with oxygen plasma to remove any contamination adsorbed by the silicon wafer surface. After being exposed to oxygen plasma, the silicon wafer is stuck to a non-residue scotch tape, with the bulk crystals of the relevant two-dimensional material stuck to the tape's sticky side. The substrate (with tape) is annealed for 2–5minutes at 100°C to complete the exfoliation process, as opposed to removing the tape right away. The substrate-tape sample is allowed to cool to room temperature (20 to 25 degrees Celsius) following the annealing process. Then, the scotch tape is peeled off slowly and carefully to deposit high-quality, large-area 2D flakes with higher deposition yields [20]. Due to the creation of a pressure differential and an increase in interfacial tension, the additional annealing step added to the current process increases the van der Waals interaction between the silicon wafer and the uppermost layer of the crystal. Interaction and reduction of the total interaction range between the silicon wafer surface and the outermost layer of the 2D crystal.

Large-area monolayer and few-layer 2D flakes are successfully transferred from their bulk counterparts when these parameters interact appropriately [19,20].

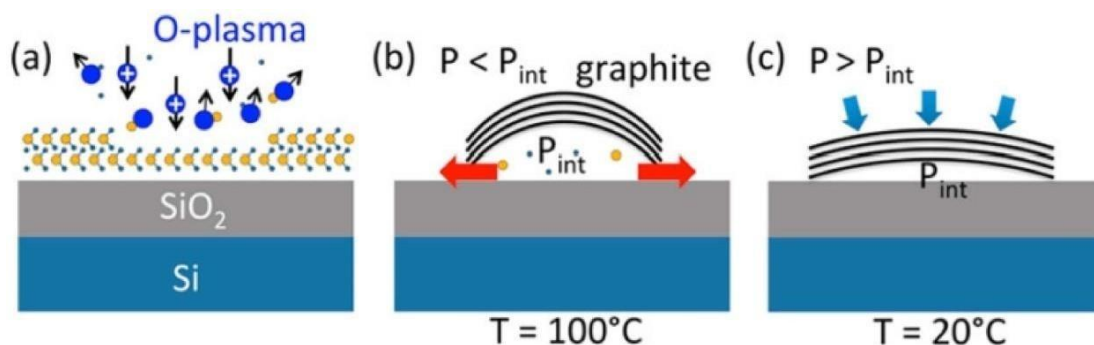


Figure 7: Schematic diagram of MHE process. (a) Cleaning the substrate by oxygen plasma. (b) Annealing of the substrate contacting graphite on adhesive tape. (c) Cooling to room temperature and a now tightened boundary result in a reduction of the pressure at the interface [20].



Figure 8: Optical image of MoS₂ by using Micromechanical Hot Exfoliation

Chapter-3

Characterization Techniques

After the deposition of 2D material on substrate (glass/Si/SiO₂), our next essential task is to characterize the material, so that we can know its optical, electronic, and physical properties. Common methods include, optical microscopy especially for preliminary analysis of materials, Raman spectroscopy, for vibrational and structural analysis, Photoluminescence (PL) spectroscopy used for electronic and optical properties of materials and atomic force microscopy (AFM) often used for surface characterization. We will discuss these techniques one by one.

3.1 Optical Microscopy

Optical microscopy is a widely used technique that employs visible light and a system of lenses to magnify small objects, allowing for detailed visualization of structures that are otherwise invisible to the naked eye. It was discovered in the late 16th to early 17th century. It is used in biology, materials science, and medicine for examining cells, tissues, and microstructures. In optical microscopy we used a high-resolution optical microscope to see graphene and some other 2D material flakes [23].

$$C = \frac{I_{\text{sample}} - I_{\text{background}}}{I_{\text{background}}}$$

Where “C” is optical constant, which determine the number of layers of 2D flakes which is deposit on the substrate. Here “I” is the intensity of light which is reflect or transmit from the deposit 2D flakes.

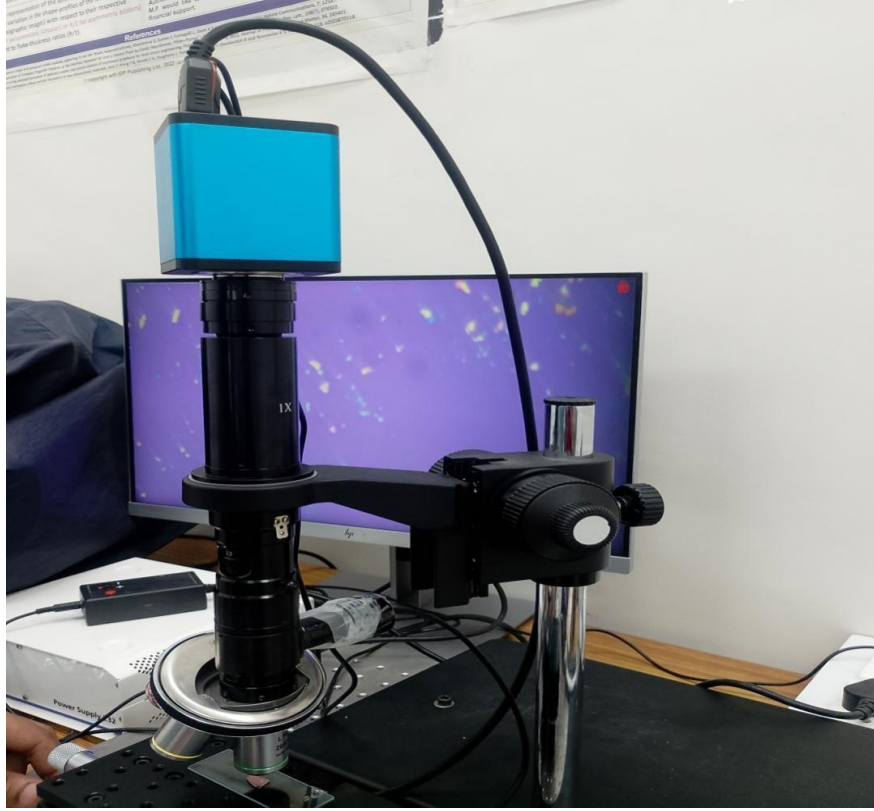


Figure 8: Represent the instrument image of Optical microscopy.

3.2 Raman Spectroscopy

Raman spectroscopy technique is used to characterize 2D material flakes. It was discovered in 1930 by a famous Indian scientist C.V. Raman. This technique is based on inelastic scattering of light. When monochromatic laser light shine onto a system of interest, its frequencies get split into higher and lower frequencies [24]. The shift in frequencies of photons are due transitions between various rotational and vibrational states of the molecule. While rotational states are difficult to absorb in non-gaseous materials due to their small nature, and thus majority of Raman spectrum reflect the vibrational transition of molecules [26]. The intensity of Raman response is proportional to the change in the electric dipole-electric dipole polarizability induced by changes in the vibrational and rotational states, and thus only materials with electric dipole-electric dipole polarizability can produce a Raman response. For the case of graphene there are two high frequency modes, G mode and 2D mode, which are found at around 1580 cm^{-1} and 2675 cm^{-1} , respectively, are included in the Raman spectrum of defect-free graphene. The G mode is connected to the in-plane vibrations of the Sp^2 carbon atoms [24] and is a first-order process that involves a single phonon (-point phonon with E_{2g}

symmetry) [25]. Conversely, the 2D mode is linked to the out-of-plane vibrations in multilayer graphene [24] and is a second order process involving two phonons (with A₁ symmetry) [25]. Consequently, the number of layers found in the graphene flakes determines the intensity of the 2D wave. When the number of layers grows, the intensity of the 2D peak decreases and becomes lower for multi-layer graphene and equal to the G peak for bilayer graphene. The intensity of the 2D peak is highest for monolayer graphene (higher than the G peak) [Ref. fig 11].

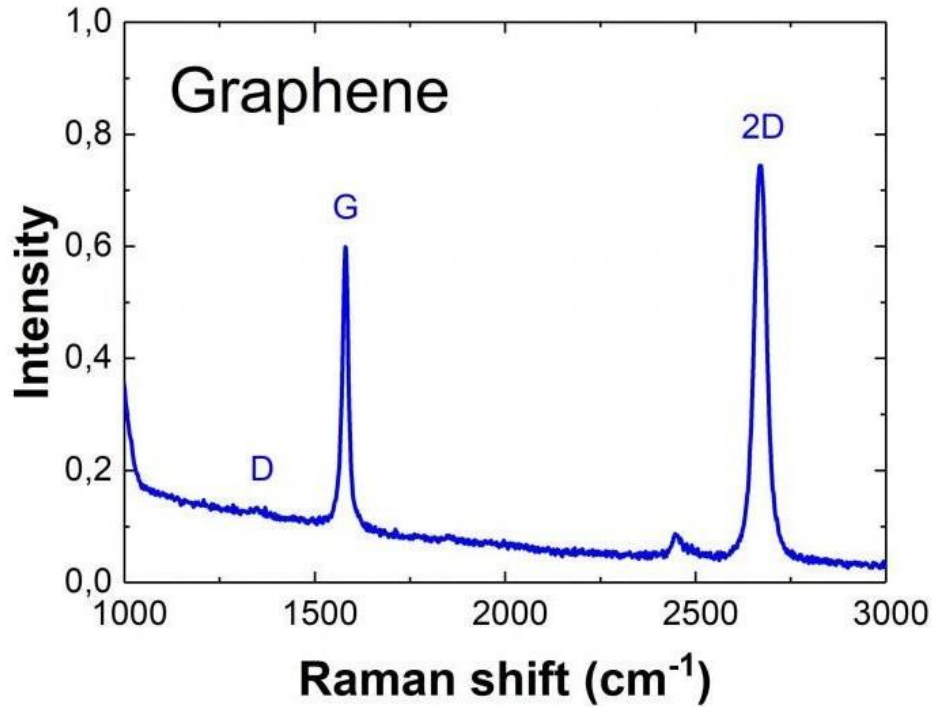


Figure 9: Raman peak of graphene for monolayer.

With the help of Raman spectra, we can determine number of layers deposited onto the substrate, also we can find defect on the flakes by seeing “D” peak which is at around 1340cm⁻¹ for graphene. The number of layers in deposited 2D Transition Metal Dichalcogenides (TMDCs) may be determined using the E₁ 2g and A₁g Raman modes of 2D TMDCs, just like in graphene [24]. The M and X atoms’ in-plane vibrations are linked to the E₁ 2g mode, whereas the X atoms’ out-of-plane vibrations are linked to the A₁g mode. The frequency of the E₁ 2g mode reduces (red shifts) as the number of layers in a 2D TMDC goes from monolayer to multilayer, whereas the frequency of the A₁g mode increases (blue shifts) in

addition, it is discovered that the two modes' FWHMs depend on the TMDC's flakes' thickness and decrease as the flakes thickness increases [24].

Chapter-4

Transfer techniques

4.1 Dry Transfer

The dry transfer method is a widely used technique for transferring two-dimensional (2D) materials, such as graphene, transition metal dichalcogenides (TMDs), or hexagonal boron nitride (h-BN), onto desired substrates without the need for wet chemicals. It was suggested for large-scale transfer process using the adhesion difference in transfer film/2D material and 2D material/target substrate interfaces. In other words, the adhesion between 2D material and a target substrate should be greater than that between 2D material and a transfer film for transferring 2D material without damages [32]. This adhesion difference support 2D material from damage by external force [33].

There are some crucial steps of this transfer process [[32], [35]].

1. Exfoliate the bulk crystal by using tape.
2. Transfer the material's 2D flakes, onto a viscoelastic polymeric film, like polydimethylsiloxane (PDMS) film and Poly propylene carbonate (PPC). It is possible to locate and identify the necessary 2D flake by looking at the sample using an optical microscope.
3. In a fulcrum configuration, the PDMS and PPC stamp is attached to a glass slide that is connected to a micro-manipulator.
4. By coordinating the film-attached 2D material with the target substrate and bringing the two into contact, and pick the 2D flake from donor substrate with the help of PDMS and PPC.
5. After that, picked 2D flakes are gradually transferred to the target substrate by hot pressing method.

(Note: Initially PDMS is used as a transfer film, but there is a problem that 2D materials cannot come into contact with PDMS uniformly with the surface of 2D material and donor substrate due to ripple pattern, so eradicate this problem we use PPC with PDMS).

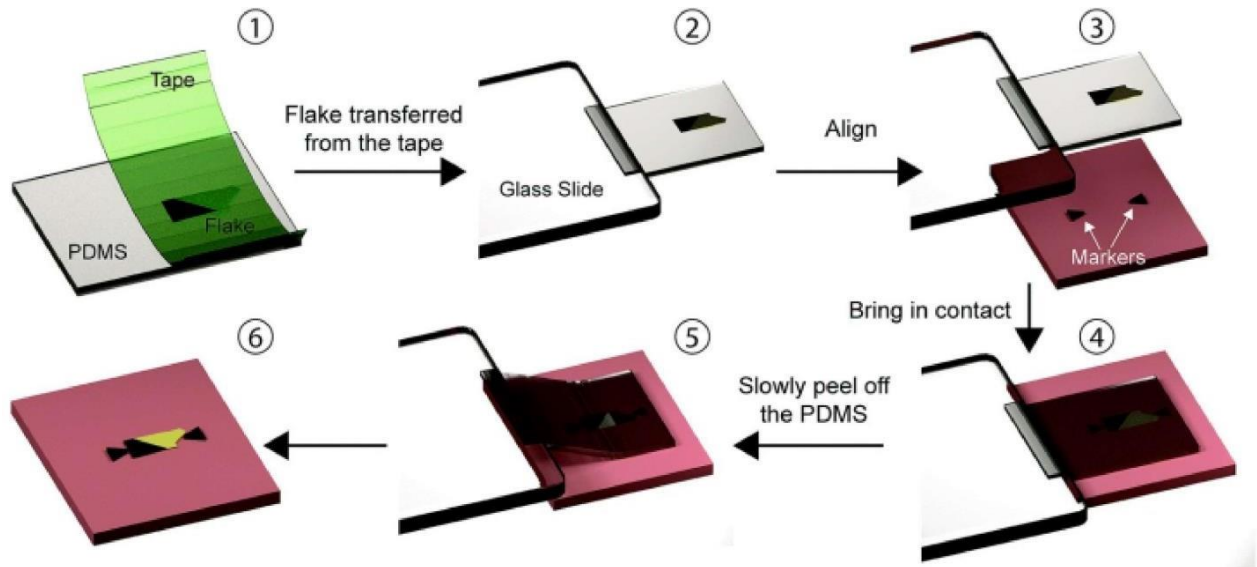


Figure 10: This figure corresponds to Dry transfer method [32].

4.2 Wet Transfer

This method is also widely used in nanotechnology and materials science to preserve the integrity of 2D layers while avoiding contamination or mechanical damage. This approach was initially used by G.F. Schneider et al. [34] to deposit graphene flakes onto patterned substrates.

There are few crucial steps of this transfer process [[32], [35]].

1. Exfoliate the bulk crystal on Hydrophilic substrate (SiO_2).
2. Coat 2D material-substrate system with hydrophobic polymer like polymethyl methacrylate (PMMA). Between substrate and PMMA 2D flakes are there.
3. Dip PMMA, 2D flakes and donor substrate stack in an etchant to etch away the donor substrate.
4. After this process PMMA and 2D flakes float on the deionized water (DI) for rinsing residual etchant.
5. Now, this PMMA and 2D flakes is scooped by target substrate. And dipped into a solvent (e.g., acetone) for removing PMMA.

However, there are many methods involve in wet transfer process, as we discuss above is one of them [32].

The wet transfer method is advantageous for its ability to handle large-area films and maintain high material quality, but it requires precise control to avoid wrinkles, tears, or contamination.

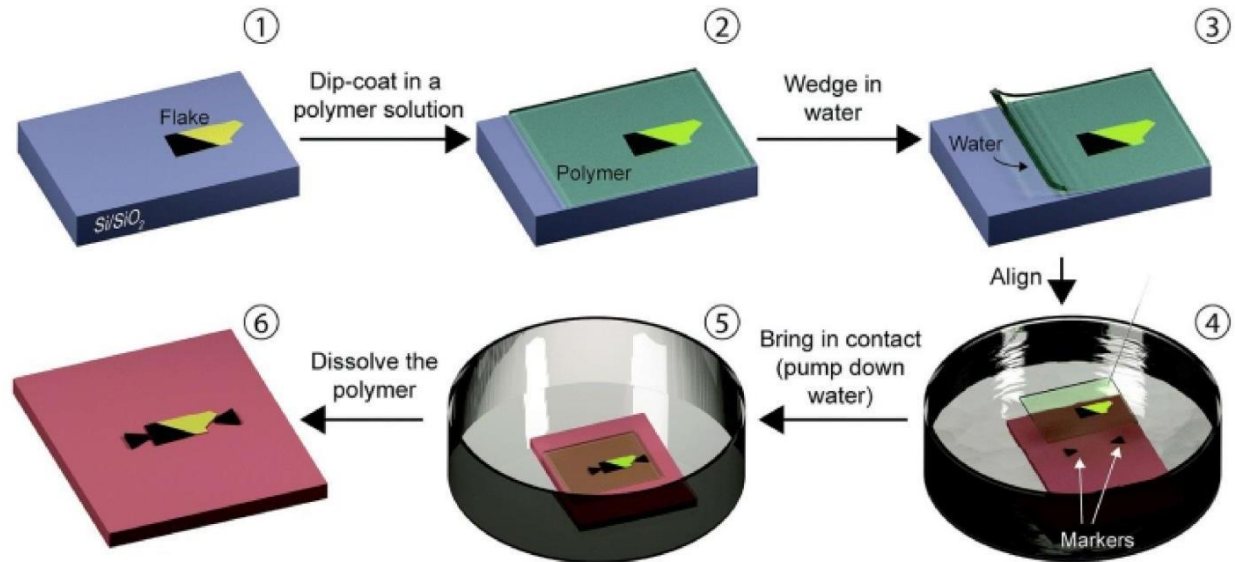


Figure 11: This figure corresponds to Wet transfer technique [32].

Chapter-5

Characterization and Results

In this chapter, I present the characterization and analysis of the graphene samples synthesized using the chemical vapor deposition (CVD) technique and subsequently transferred using the PMMA-assisted wet transfer method. The aim of this section is to evaluate the quality, uniformity, and structural properties of the grown graphene through various characterization tools such as optical microscopy and Raman spectroscopy. These techniques provide essential insights into layer thickness, defect density, crystalline quality, and the overall effectiveness of the growth and transfer processes. The following results summarize the observations obtained from the optical and Raman studies of the CVD-grown graphene placed on SiO₂/Si substrates.

5.1 Optical Characterization of graphene

Optical microscopy was used to assess the uniformity, coverage, and approximate layer thickness of the transferred graphene based on color contrast on SiO₂/Si substrates. In the PMMA-transferred sample, region A shows a uniform orange contrast, which is characteristic of bilayer graphene and confirms that the transfer was successful in those areas. However, region B does not exhibit this contrast, indicating that portions of the graphene layer are missing or were partially removed during the copper etching process.

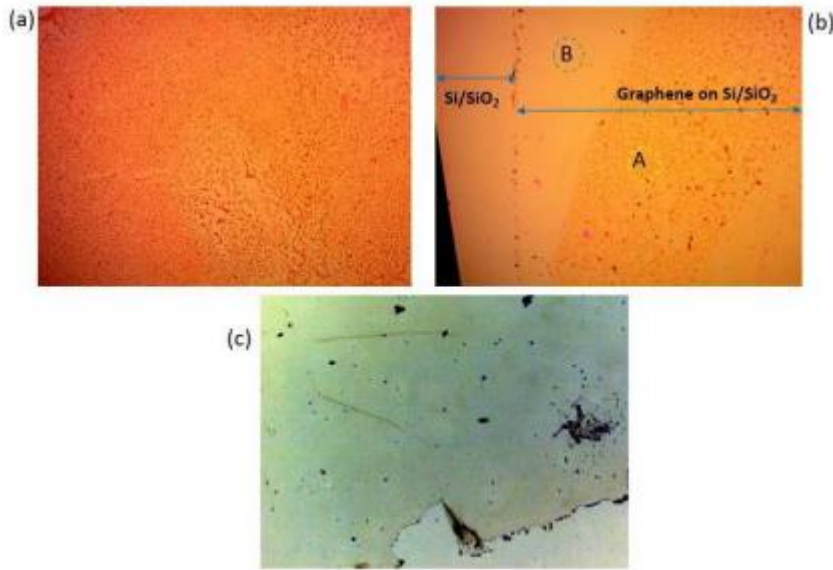


Figure 13: Optical microscope images with 5x magnification at 50.0 μ m scale of (a) transferred fast-cooling graphene by PET/silicone and (b) by PMMA materials onto native oxide SiO₂(300nm)/Si substrate (c) Graphene easy release CVD graphene on SiO₂(10nm)/ α Ge(650nm)/NiCr(0.7nm)/Au(125nm)/glass substrate

This incomplete coverage is linked to the prolonged etching time required for PMMA-supported Cu foil, which took nearly 23 hours to dissolve fully in FeCl_3 , increasing the likelihood of underlayer graphene being unintentionally etched away. The contrast difference between regions A and B highlights how PMMA transfer can yield good-quality bilayer graphene, but is highly sensitive to etching duration and process control. Overall, the optical images confirm the presence of bilayer graphene in certain regions while also revealing areas of incomplete transfer due to extended etching.

5.2 Raman Characterization of Graphene

Raman spectroscopy is one of the most powerful and widely adopted techniques for the structural characterization of graphene. It provides rapid, non-destructive information about the number of layers, defect density, strain effects, and overall crystalline quality of the transferred graphene films. The key Raman features used for graphene analysis include the D band ($\sim 1350 \text{ cm}^{-1}$), the G band ($\sim 1580\text{--}1600 \text{ cm}^{-1}$), and the 2D band ($\sim 2680\text{--}2750 \text{ cm}^{-1}$). Each of these peaks carries important information regarding the physical condition of the graphene sheet. The D peak is associated with disorder and lattice imperfections; the G peak corresponds to the in-plane stretching mode of sp^2 -bonded carbon atoms; and the 2D peak originates from a second-order two-phonon scattering process that is highly sensitive to the number of layers and stacking order.

In this study, Raman spectroscopy was employed to assess the quality of graphene transferred onto SiO_2/Si substrates using PMMA as the support material. The fast-cooling transferred graphene shows distinct D, G, and 2D peaks confirming successful transfer and preservation of the graphene lattice. The D peak appears at approximately 977 cm^{-1} with very low intensity, indicating a low defect concentration and minimal structural disorder. The G peak observed at $\sim 1592 \text{ cm}^{-1}$ reflects the presence of well-ordered sp^2 carbon networks, with only minor shifts that may arise from slight strain

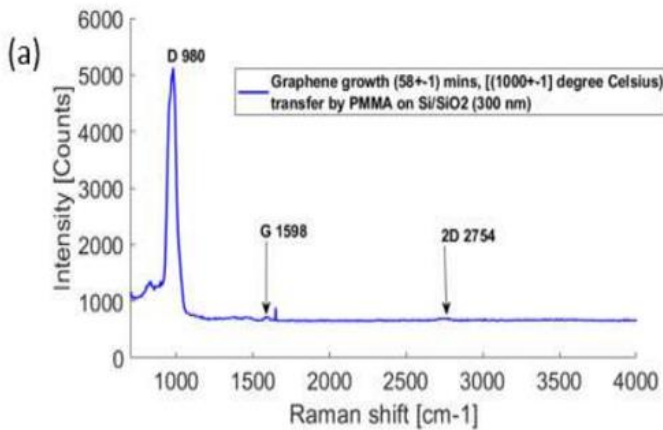


Figure 13: Represent Raman peak for graphene

or doping introduced during the transfer process. The 2D peak, located around 2728 cm^{-1} , is sharp and relatively intense, demonstrating that the crystalline structure of graphene is retained after growth and transfer.

The ratio of the 2D to G peak intensities (I_{2D}/I_G) is a crucial parameter used for layer identification. In the case of fast-cooled PMMA-transferred graphene, the intensity ratio is approximately 1, which is characteristic of **bilayer graphene**. This observation aligns with the expected outcome of the growth parameters, specifically the methane flow rate used during the CVD process. The well-defined 2D peak, combined with a low D peak and a stable G peak, verifies that fast cooling contributes significantly to maintaining the bilayer stacking and reducing defect generation.

Overall, the Raman analysis confirms that the combination of CVD growth conditions and PMMA-assisted fast cooling results in high-quality bilayer graphene. The clear Raman fingerprints and low defect signals indicate that the graphene remains structurally intact and is suitable for subsequent device fabrication and advanced characterization.

5.3 Conclusion

In this study, we successfully synthesized graphene using the chemical vapor deposition (CVD) technique and evaluated its quality through Raman spectroscopy and optical microscopy. Raman analysis confirmed the formation of **bilayer graphene** under fast-cooling conditions, indicated by a clear G peak, a sharp 2D peak, and an intensity ratio $I_{2D}/I_G \approx 1$, along with a very low D peak reflecting minimal defects. Optical microscopy further validated these findings by showing uniform orange contrast regions on SiO_2/Si substrates, characteristic of bilayer graphene. The comparison of transfer methods showed that **PMMA-assisted transfer preserves bilayer regions**, although extended copper etching can cause partial graphene loss, while PET/silicone transfer yielded more uniform coverage due to faster and more stable Cu etching. Overall, the characterization results demonstrate that growth conditions, cooling rate, and transfer technique significantly influence graphene quality. Continued optimization of substrate preparation, annealing, and transfer parameters will further enhance the uniformity and scalability of CVD-grown graphene for practical device applications.

Chapter-6

Summary

This study focuses on the growth, transfer, and characterization of CVD-grown graphene using copper substrates and PMMA-assisted transfer techniques. The work examines how different stages of the CVD process—such as copper surface preparation, high-temperature annealing, controlled methane flow, and cooling rate—directly influence the quality and number of graphene layers. Raman spectroscopy was used as a primary characterization method to evaluate the structural quality, defect levels, and layer thickness of the transferred graphene. The fast-cooling PMMA-transfer samples showed clear D, G, and 2D peaks, with an I_{2D}/I_G ratio of approximately 1, confirming the formation of bilayer graphene with relatively low defect density. Optical microscopy further supported these results by revealing bilayer contrast regions on SiO₂/Si substrates, though certain areas showed incomplete coverage due to prolonged Cu etching required during PMMA transfer. Overall, the work demonstrates that optimized CVD growth conditions combined with PMMA transfer can successfully produce bilayer graphene. However, challenges such as long etching times, underlayer loss, and transfer-induced defects highlight the need for improved processing routes to achieve more uniform and scalable graphene films for future device applications.

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